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Subject : Problem Solving Using Python

Batch : 03

PROJECT: HANGMAN GAME

AIM

To develop a Python-based Hangman game that allows users to guess words from different technical categories with features like difficulty levels, hints, and scoring.

ALGORITHM

1. Start the program
2. Define categories with words
3. Randomly select a category and word
4. Ask user to choose difficulty (easy/medium/hard)
5. Assign attempts based on difficulty
6. Display category and word length
7. Initialize empty guessed letters set
8. Repeat until attempts = 0:
    - Display current word progress
    - Take user input (letter or hint)
    - If hint:
        - Reveal a random unguessed letter
        - Reduce attempt
    - Else:
        - Validate input

        - Check if letter is correct
        - Update guessed letters
        - Reduce attempts if wrong

9. If all letters guessed:
    - Display win message
    - Calculate score
10. Else:

- Display lose message

11. Ask user to play again
12. Display final score
13. End

CODE

```
import random
import time

categories = {
   "Programming Languages": ["python", "java", "javascript", "golang"],
   "Game Titles": ["minecraft", "fortnite", "valorant", "pubg"],
   "Computer Parts": ["processor", "motherboard", "graphics", "ram"]
}

def choose_word():
   category = random.choice(list(categories.keys()))
   word = random.choice(categories[category])
   return word, category

def display_word(guessed, word):
   return " ".join([letter if letter in guessed else "_" for letter in word])

def get_attempts(level):
   if level == "easy":
      return 6
   elif level == "medium":
      return 4
```

```
    else:
        return 3

def play_hangman():
    word, category = choose_word()
    guessed_letters = set()
    hint_used = False
    score = 0

    print("\nHANGMAN GAME (Tech Edition)")
    level = input("Choose difficulty (easy/medium/hard): ").lower()
    attempts = get_attempts(level)

    print("Category:", category)
    print("Hint: Word length is", len(word))

    while attempts > 0:
        print("\nWord:", display_word(guessed_letters, word))
        print("Attempts left:", attempts)
        print("Guessed:", ", ".join(sorted(guessed_letters)) if guessed_letters else "None")

        choice = input("Enter letter or type 'hint': ").lower()

        if choice == "hint":
            if not hint_used:
                remaining = [l for l in word if l not in guessed_letters]
                if remaining:
                    hint_letter = random.choice(remaining)
                    print("Hint letter:", hint_letter)
                    hint_used = True
                    attempts -= 1
                else:
                    print("No letters left to reveal!")
            else:
                print("Hint already used!")
            continue

        if len(choice) != 1 or not choice.isalpha():
            print("Enter a valid letter!")
            continue
```

```
        if choice in guessed_letters:
            print("Already guessed!")
            continue

        guessed_letters.add(choice)

        if choice not in word:
            attempts -= 1
            print("Wrong guess!")
        else:
            print("Correct!")

        if all(letter in guessed_letters for letter in word):
            score += attempts * 10
            print("\nYou WON! Word was:", word)
            print("Score gained:", score)
            return score

    print("\nYou LOST! Word was:", word)
    return 0

def main_hangman():
    total_score = 0

    while True:
        total_score += play_hangman()
        again = input("\nPlay again? (yes/no): ").lower()
        if again != "yes":
            print("\nFinal Score:", total_score)
            print("Thanks for playing!")
            break

main_hangman()
```

SAMPLE OUTPUT

```
HANGMAN GAME (Tech Edition)
Choose difficulty (easy/medium/hard): easy

Category: Game Titles
Hint: Word length is 8

Word: _ _ _ _ _ _ _ _
Attempts left: 6
Guessed: None
Enter letter: m
Correct!

Word: m _ _ _ _ _ _ _
Attempts left: 6
Guessed: m

Enter letter: z
Wrong guess!

Attempts left: 5
...
```

RESULT

The Hangman game was successfully implemented using Python.
The program allows users to guess words with interactive gameplay, difficulty levels, hints, and scoring system.